The Structure of a BARRIER REEF

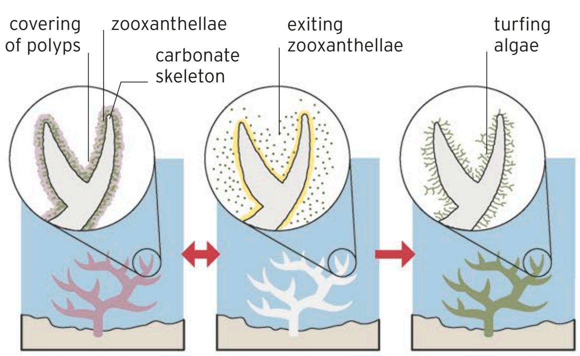
A barrier reef is a type of coral reef widely separated from a continental mainland or island by a deep channel or lagoon. One circumstance in which such a reef can develop is when a volcanic island in the tropics begins to sink (as typically occurs when its volcanic activity ceases). As the island subsides, any fringing reef hugging its shores continues to grow upward. A gradually widening lagoon then develops between the reef and the island. A second way a barrier reef can develop is on and around the edge of a continental shelf when sea level rises. This is how the world's largest barrier reef—the Great Barrier Reef off Queensland, Australia—formed.

Development of the Great Barrier Reef

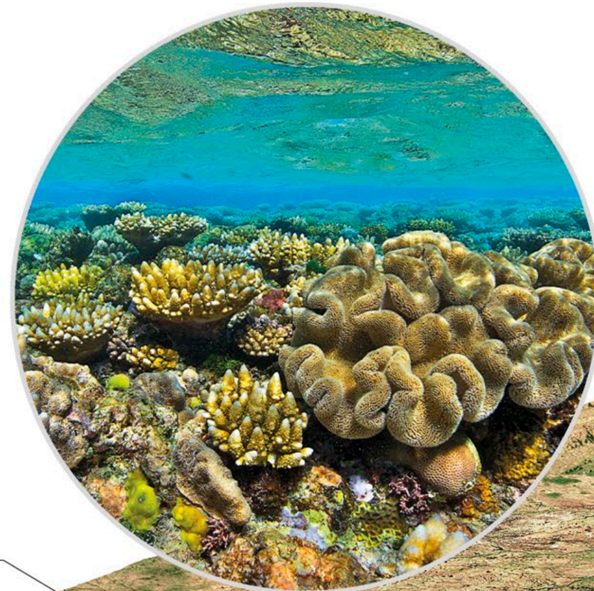
The Great Barrier Reef has taken some 20 million years overall to develop, although much of the structure visible today has grown since the last ice age on top of an older platform of continental shelf. Around 18,000 years ago, the sea was about 390 ft (120 m) lower and the outer edge of today's barrier reef was the edge of continental Australia. As sea level rose, ocean water and then corals encroached on what had been hills of the Australian coastal plain. By 13,000 years ago, many of these hills had become islands on the continental shelf. Eventually, they were submerged and corals grew over them. Sea level on the Great Barrier Reef has not changed significantly in the last 6,000 years.

CORAL BLEACHING

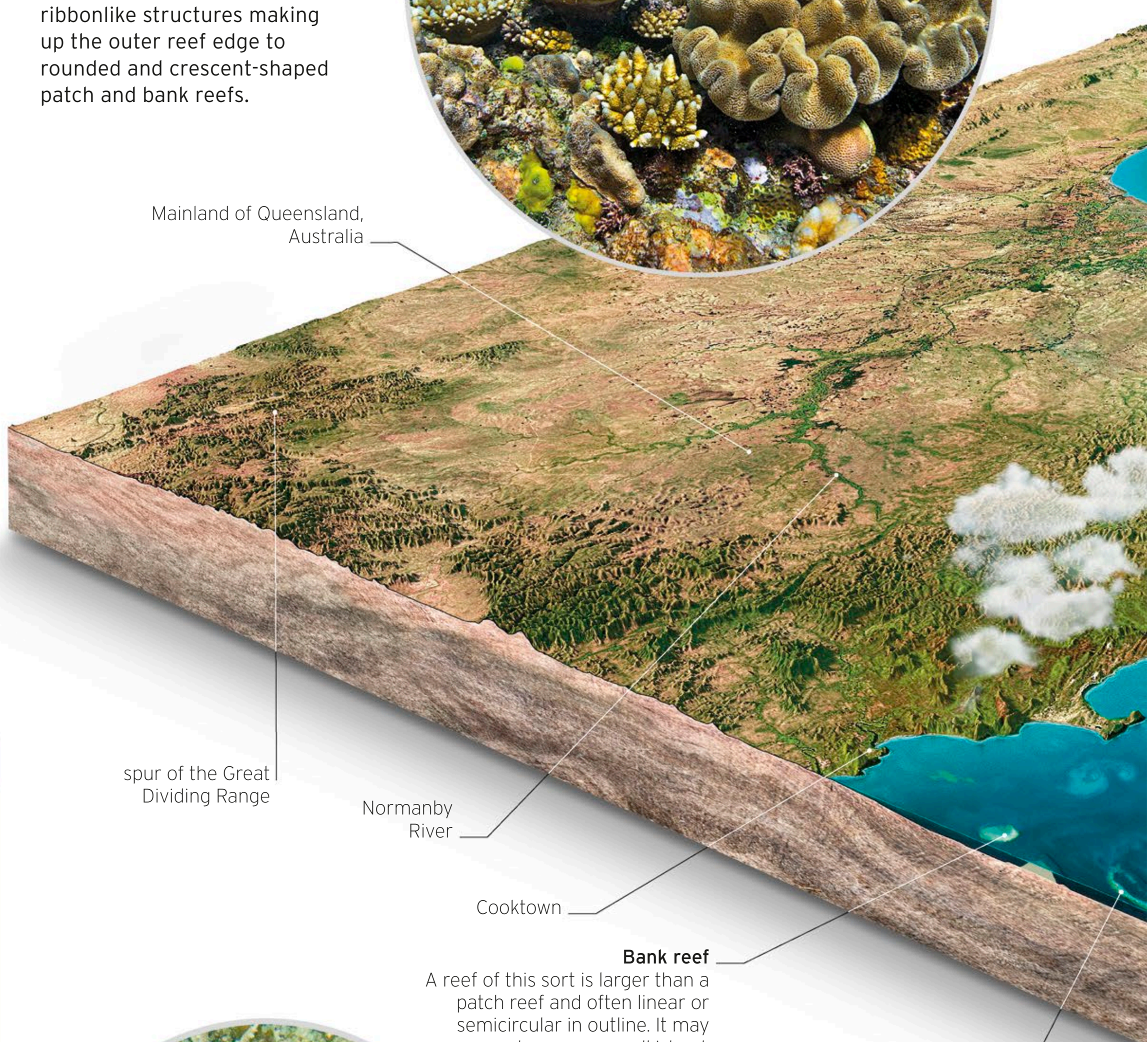
Bleaching occurs when zooxanthellae (tiny algal organisms), which give corals their colors, leave the coral polyps due to a change in seawater temperature. This is reversible if the temperature change is not too prolonged or severe. But bleached coral becomes more vulnerable to disease.
If it becomes colonized by a mat of other types of algae, called "turfing" algae, this signifies irreversible damage.



▽ REEF BUILDERS
Some polyps, called stony corals or hexacorals, secrete small amounts of calcium carbonate (limestone) as part of their body structures, building on the substrate underneath. When polyps die, their limestone skeletons remain, building up the reef.



▶ THE GREAT BARRIER REEF
A 155-mile- (250-km-) long section of the reef, off northern Queensland, shows the overall structure. There are many individual reefs in a variety of forms ranging from ribbonlike structures making up the outer reef edge to rounded and crescent-shaped patch and bank reefs.



◁ BLEACHED CORAL
A bleached coral appears white due to loss of algal organisms that supply the color. The algae provide most of the coral's energy: after they leave, the coral begins to starve and may eventually die.